Public Provision and Private Competition in Health Insurance

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Abstract

This paper studies the effect of expanding public provision of health insurance on the parallel private market. I consider an expansionof Medicaid eligibility, which shifts a particular subsetof consumers fromthe private individual market. I develop a structural model of endogenous insurer entry,capturing insurer responses to the policy, competition, and consumer sorting. Applying the model to administrative enrollment and claims data from Colorado, I find that accounting for the effect on insurer market participation reveals substantial private market welfare loss and reduced government subsidy spending concentrated insmaller markets. These effects arise primarily from increased probabilities of having no insurer or only one insurer, which reduce consumer choice and private enrollment. The estimates provide insights into recent policy proposals to extend Medicaid eligibility.

1 Introduction

Social programs often combine public benefits with private options to balance expanding access, preserving consumer choice, and limiting public expenditure. Income cutoffs are frequently used to determine who qualifies for public benefits, such as eligibility for public early-childhood programs in the United States and eligibility for public housing loans in Singapore. A central policy choice is where to set the income cutoff for public support. While extending public benefits to households with higher incomes can improve their affordability, it also changes the demand facing private providers, creating a trade-off between extending public support and sustaining competition and choice for those who continue to rely on private markets.

I study this trade-off in U.S. health insurance markets for lower-income adults, where lower-income adults below an income threshold are typically eligible for public health insurance coverage through Medicaid. The main alternative to Medicaid is the private individual market, where

consumers purchase plans directly from insurers. Because Medicaid eligibility affects both the size and the composition of private market demand, evaluating depends on quantifying how private insurers strategically respond.

Where to draw this boundary remains an active policy question. The 2010 Affordable Care Act (ACA) established an effective Medicaid expansion threshold of 138 percent of the federal poverty level (FPL), while Minnesota and New York have longstanding programs extending public coverage through 200 percent FPL. More recently, Oregon introduced such coverage in 2024, and Colorado is evaluating a similar option as of 2026. Expanding public eligibility also raises the question of whether the remaining private market can sustain insurer participation and competition. Exchange withdrawals have highlighted the importance of sufficient expected returns (GAO, 2019) and prompted state efforts to prevent counties from losing exchange coverage altogether (Lucia et al., 2017). These concerns are especially salient in markets with few insurers: in 2018, half of U.S. counties had only one exchange insurer (McDermott and Cox, 2020). However, research on Medicaid eligibility has given relatively little attention to insurer entry and exit, leaving an important channel less understood.

Motivated by these policy developments and concerns about insurer participation, I evaluate the welfare consequences of expanding Medicaid eligibility from 138 to 200 percent FPL using Colorado enrollment and claims data from 2017–2019. Colorado provides a useful setting for this analysis, with direct relevance to the state's ongoing consideration of broader public coverage. Its Medicaid program pays for most physical health services through a publicly administered fee-for-service system. This structure limits the scope for individual-market insurers to offset enrollment losses by serving newly eligible consumers through Medicaid plans, making Colorado particularly useful for studying how expanding public coverage affects participation in the remaining private market. Colorado's counties also vary substantially in population, enrollee composition, medical costs, and insurer participation. Combined with detailed enrollment and claims data, this variation informs the estimation of demand, costs, and entry decisions and allows me to examine how the consequences of expansion differ across markets with different local characteristics.

I begin by documenting the market characteristics that are key to the main welfare conclusions of this paper. First, I show that the expansion population accounts for a sizable share of the existing private enrollment, suggesting that the expansion would remove a substantial share of

insurers' existing profit. I then show that subsidies substantially narrow geographic differences in the prices; in particular, net premiums are lowest in the smallest markets despite their higher gross premiums. Finally, I show that existing insurer participation is much sparser in smaller markets, leaving smaller markets more exposed to being left with a monopoly or even losing their last insurer, motivating an analysis that accounts for insurers' market participation decisions.

To quantify the welfare and government spending consequences of extending Medicaid eligibility, I build and estimate a model of insurance demand and supply with insurer market participation as a central margin of adjustment. On the demand side, individuals with different demographics and health states choose among available insurance plans differentiated by premiums, coverage, and networks. I model consumer utility for networks following Ho and Lee (2017) and nest a hospital demand model within the insurance demand model to capture local network differences between insurers, as well as how consumers with different anticipated health needs sort between insurers.

On the supply side, insurers decide what premiums to charge and which markets to serve. The model allows changes in public insurance eligibility to affect private-market outcomes through both the intensive margin of pricing and the extensive margin of insurer participation. I model entry as a static game of incomplete information in which insurers participate in a market when expected operating profits exceed their privately observed fixed costs. The incomplete information formulation ensures that an entry equilibrium exists despite the potentially nonmonotonic effects of competition on insurer profits under risk selection. I then estimate the entry-cost distribution using a two-step approach analogous to Pakes, Ostrovsky, and Berry (2007). This approach uses the equilibrium behavior observed in the data and avoids repeatedly solving for entry equilibria and choosing among potentially multiple equilibria during estimation.

The resulting equilibrium participation probabilities allow me to quantify how expanding public eligibility changes the likelihood of different market structures. The counterfactual results reveal that expanding public coverage has sharply different consequences across local markets, despite little change in average consumer welfare. Among the population remaining in the individual market, consumer surplus falls by just \$0.03 per person per month statewide. This average masks losses of \$15.01 in the smallest third of counties and \$5.03 in the middle third, compared with a gain of \$0.59 in the largest third.

These geographic differences emerge primarily through insurers' participation responses: when participation probabilities are held fixed at baseline, welfare changes are close to zero in every market-size group. Accounting for the policy's effect on insurer participation is important because the premium subsidies adjust with insurance prices, cushioning consumers against increases in gross premiums and limiting the welfare effects of pricing responses. Holding subsidies' reference prices fixed, insurers' pricing responses increase consumer surplus, but allowing those reference prices to adjust offsetts approximately almost all of the initial benefit.

Moreover, different market structures imply different enrollment, consumer surplus, and subsidy spending. While the change in participation is similar across markets, smaller markets begin with fewer insurers. As a result, participation responses that shift markets toward market structures with lower consumer surplus are most pronounced in smaller markets. Even though expected participation falls by approximately 0.3 insurers in each market-size group, the probability that no insurer participates nearly doubles from 7.7 to 15.0 percent, and the probability of monopoly increases from 42.2 to 49.0 percent in the smallest markets, while larger markets generally retain multiple insurers.

Insurer participation also changes the interpretation of government spending reductions. Statewide, spending on premium subsidies falls more when participation is held fixed relative to when participation is allowed to adjust. On the other hand, participation responses deepen the reductions in smaller markets. In these markets, a greater risk of losing insurance availability lowers enrollment and subsidy payments, and spending reductions are a result of diminished access and consumer welfare losses.

This analysis highlights insurer participation as a key link between public eligibility rules and the welfare of consumers who remain in private markets. While the estimates are specific to the expansion of Medicaid eligibility considered here, the underlying mechanisms are relevant to policies that shift consumers between public programs and subsidized private alternatives more generally.

Relation to Literature This paper contributes to the literature on how public insurance expansions affect private insurance markets. Cutler and Gruber (1996) and Gruber and Simon (2008) study whether expansions in Medicaid eligibility crowd out private coverage demand. Clemens

(2015), Peng (2017), and Sen and DeLeire (2018) examine how Medicaid expansion changes the composition and premiums of the private individual market. I extend this literature by examining how expansion affects coverage options and welfare for consumers who remain ineligible for public insurance. Closest to this paper, Kanimian (2025) studies whether Medicaid eligibility thresholds should reflect geographic differences in the cost of living, allowing Marketplace insurers to adjust premiums and Medicaid plans to adjust quality while holding insurer participation fixed. I instead study a uniform eligibility expansion with endogenous private insurer entry and exit. This allows me to quantify an additional consequence of expanding public coverage: the loss of private coverage options for consumers who do not gain access to the public program.

This paper also connects to research on market segmentation in health insurance. Dickstein, Ho, and Mark (2024) study pooling the small-group and individual markets, allowing insurers to reprice while holding plan offerings fixed. Fleitas et al. (2025) examine policies that move households from the small-group market into the Marketplace. These papers highlight how the allocation of consumers across separately regulated insurance pools shapes selection and welfare. I examine this question at the Medicaid–individual market boundary, allowing changes in the allocation of consumers to alter which insurers participate in the remaining private market. My analysis highlights how an eligibility reform could change not only the terms of available coverage but also whether that coverage remains available.

Finally, this paper relates to the literature on how selection, market power, and regulation shape insurance markets (Starc, 2014; Ericson and Starc, 2015; Mahoney and Weyl, 2017; Jaffe and Shepard, 2020; Shepard, 2022; Tebaldi, 2025). Within this literature, this paper joins a small group of recent papers that focus on insurer participation as a key channel affecting market outcomes. Geddes (2025) studies how geographic pricing and service-area rules affect entry and coverage availability, while Zahn (2025) examines how subsidies affect selection between public and private Medicare, insurer entry, and product offerings. While they investigate how pricing and subsidy rules in a given market segment shape insurers' incentives to serve a market, I study a change in the boundary between public and private coverage, holding the private market's regulatory framework fixed. Methodologically, this paper formulates the entry equilibrium as an incomplete information game, while Geddes and Zahn both model the entry equilibrium as one of complete

information.[1]

The rest of this paper is organized as follows. Section 2 describes the institutional background. Section 3 describes the data used for this paper and shows descriptive facts that illustrate the market characteristics driving the main welfare conclusions. Section 5 presents the model and describes the estimation procedure and results. Section 6 simulates and interprets the counterfactual outcomes. Section 7 concludes.

2 Institutional Background

Medicaid Medicaid is the largest source of funding for medical and health-related services for people with low income in the United States, including children, pregnant individuals, older adults, and people with disabilities. As the nation's largest public health insurance program, it covers over 80 million individuals—roughly one in four Americans. As the nation's largest public health insurance program, it covers over 80 million individuals—roughly one in four Americans.

Each state administers their Medicaid programs subject to federal requirements. Colorado's Medicaid program, Health First Colorado, covers over one million adults and children each year. Eligibility is primarily income-based. In 2014, Colorado implemented its Medicaid expansion program to extended coverage to nearly all non-elderly adults with household incomes up to an effective threshold of 138% FPL of the Federal Poverty Level. Medicaid enrollees in Colorado receive comprehensive benefits including physical, behavioral, and dental health services alongside long-term supports. Enrollees pay zero premiums and pay little, if anything, out-of-pocket, for most services. Eligibility for Medicaid generally precludes eligibility for government subsidies towards Marketplace coverage. Operationally, most health services are reimbursed fee-for-service at state-determined rates, with regional organizations coordinating care and maintaining provider networks. [2]

The Individual Market The U.S. individual health insurance market provides coverage for people who do not obtain health insurance through an employer or a public program such as Medi-

[1] Kong, Layton, and Shepard (2024) also study insurer participation and show how competition for healthier consumers can undermine entry but abstract from entry cost estimation.

[2] This applies to physical health services and long-term care services. For behvaioral health services, the regional organizations receive a fixed, monthly per-member budget and negotiate independently with local mental health providers to detemine the reimbursement fee schedule.

care or Medicaid. As of 2025, the individual market covers approximately 27 million people, which is about 8.6% of the insured population.

Colorado operates its own ACA marketplace, Connect for Health Colorado, through which consumers compare and purchase private plans. Consumers can also purchase individual coverage off the Marketplace, but federal premium tax credits and cost-sharing assistance are available only through the Marketplace. My analysis covers both channels during 2017–2019.

Plans offered in the individual market are standardized in several ways. All plans are mandated to cover essential health benefits such as preventive services and mental health benefits. The cost-sharing features of the plans, such as deductibles and coinsurance, are summarized by different metal levels, so that bronze plans cover roughly 60% of medical costs, silver plans cover 70%, gold plans cover 80%, and platinum plans cover 90%. During my sample period, insurers in Colorado are required to offer silver and gold plans, and no insurer offers platinum plans.

Individual market insurers are also subject to federal and state pricing regulations that constrain how premiums are set. Under community rating rules, premiums cannot vary based on individual health status or prior claims experience, and can only vary within limits based on factors such as age and geography. Specifically, premiums for the same plan are only allowed to vary by age according to a statutory schedule within a rating area, which are bundles of one or more counties determined by the state. Insurers must also satisfy medical loss ratio (MLR) requirements, which mandate that a minimum share of premium revenue be spent on medical claims. This restriction provides an additional constraint on insurers' pricing responses as enrolle costs change.

Insurer entry in the individual market is typically determined at the county level and is made on a plan-year basis. Insurers make annual decisions to enter, exit, or continue participation in each market several months prior to the start of the coverage year. Therefore, insurance availability can differ between counties and between years in the same county.

During my sample period, eligible households with incomes up to 400 percent FPL could receive premium tax credits. The credit covers the positive difference between the premium of the second-lowest-cost Silver plan available to the household and an income-based required contribution, capped at the premium of the selected plan. Because subsidies adjust with this benchmark, changes in gross premiums need not translate into comparable changes in consumers' net pay-

ments. Eligible households with incomes up to 250 percent FPL also receive cost-sharing reductions when enrolling in Silver plans, increasing their coverage generosity. These rules affect both consumers' plan choices and how changes in insurers' premiums and participation translate into consumer welfare.

Households can purchase insurance coverage in the individual market either through a marketplace created under the ACA or through an off-marketplace channel, but households eligible for federal financial assistance must enroll through the Marketplace to receive those benefits. The primary form of assistance is the premium tax credit (PTC), which reduces monthly premium payments for eligible households. During my sample period, eligible households with incomes up to 400 percent FPL could receive premium tax credits.[3] The credit covers the positive difference between the premium of the second-cheapest Silver plan available to the household and an income-based required contribution, capped at the premium of the selected plan. Because subsidies adjust with this benchmark, changes in gross premiums need not translate into comparable changes in consumers' net payments.

In addition to premium subsidies, lower-income households enrolling in Silver plans may qualify for cost-sharing reductions (CSRs), which lower out-of-pocket expenses. Unlike premium tax credits, which reduce monthly premiums, CSRs increase the actuarial value of eligible Silver plans by requiring insurers to offer plan variants with lower out-of-pocket costs in a way that satisfies a federally specified actuarial value. During my study period in Colorado, Marketplace enrollees with incomes between 138% and 150% of the federal poverty level qualify for Silver plans with a 94% actuarial value, while those with incomes between 150% and 200% and between 200% and 250% of the federal poverty level qualified for plans with actuarial values of 87% and 73%, respectively.

3 Data and Descriptive Patterns

3.1 Data

The main dataset used in this paper is the Colorado All-payer Claims Data (CO APCD), an administrative dataset that contains the near-universe of insurance enrollment and claims for residents

[3] Under the premium tax credit schedule applicable during my sample period, households at approximately 138% of the federal poverty level were required to contribute about 3.2% of income toward the benchmark premium, with the required contribution increasing to about 6.3% at 200% FPL and reaching about 9.5% at 300–400% FPL.

in Colorado. For each member, the dataset includes information on their choice of insurance plan, the start and end dates of coverage associated with the plan, whether the purchase was made through the ACA exchange, as well as the member's age, gender, race, and location of residence (5-digit zip code). For each claim record, the data include information on the procedure(s) filed, the amount paid by the patient and the insurer, and the facility or provider associated with the reported service.

The insured analysis population is constructed as repeated March cross-sections of the Colorado individual market from the CO APCD for 2017–2019. The analysis sample is restricted to 543,431 member-years between 18 and 64 who were enrolled on March 1 in non-grandfathered individual-market coverage.[4] Of these, I further restrict to 535,575 member-years with a unique nonmissing zip-age-sex combination. Finally, 163 member-year records with more than one insurer-metal record are excluded, leaving a final sample of 535,412 member-years.

The CO APCD does not report income, which determines eligibility for public coverage and premium subsidies. Although exchange applications collect income information to determine financial assistance, they do not provide a common income measure covering off-exchange enrollees and uninsured adults. To retain the detailed enrollment and health information in the claims while preserving uncertainty about individual income, I combine administrative records with external income distributions, assigning each insured enrollee probabilities over six income categories conditional on coverage status, county, age, sex, and observed morbidity.[5]

The assignment combines coverage-specific statewide income distributions with assumptions about geographic and health-related variation. Statewide income shares are anchored to Colorado exchange enrollment reports for on-exchange enrollees and the CPS ASEC for off-exchange enrollees. The assignment relies on two main assumptions. First, I assume that relative income differences across county-age groups measured in the ACS apply within each coverage group.

[4]The open enrollment period for the Colorado exchange ends in January of each year, and coverage typically begins in February at the latest. I use March enrollment data because some insurers appear to have enrollment data reporting lags in February. I treat March enrollment as a representative of enrollment year-round. While enrollment patterns could differ between months, incorporating enrollment timing decisions requires substantially more data and modeling and is outside of the scope of analysis for this paper (see Diamond et. al, 2026).

[5]The categories are 0–137, 138–199, 200–249, 250–299, 300–399, and 400 percent or more of FPL. They isolate the population newly eligible under the expansion and retain selected thresholds relevant to Medicaid eligibility and marketplace assistance during 2017–2019. In particular, 200, 250, and 300 percent mark boundaries in the premium-contribution schedule, and 400 percent marks the historical upper income limit for premium tax credits. The categories also broadly follow marketplace enrollment reporting bands; differences in source-data classifications are reconciled as described in Appendix A.

This allows local income composition to vary while preserving differences in statewide income distributions between on- and off-exchange enrollees. Second, I assume that the morbidity distribution conditional on broad income group, sex, and coverage status measured in the MEPS applies to Colorado. Because MEPS jointly observes income and health, it supplies information about their joint distribution that cannot be recovered from claims alone, and estimating this relationship separately by coverage status allows the income–health relationship to differ across insurance groups.

I augment the CO APCD with a sample of uninsured individuals by constructing a synthetic uninsured population with respect to the insured population. To include uninsured adults, I apply ACS uninsured-to-direct-purchase coverage ratios to the corresponding county-age cells in the CO APCD sample. Because neither income nor morbidity is observed for these individuals, I assign a joint distribution using the same geographic adjustment and the MEPS relationship for uninsured respondents, calibrated to their statewide ACS income distribution. Appendix A provides the construction details.

3.2 Descriptive Patterns

This section presents descriptive statistics on potential exposure to the Medicaid eligibility expansion, price dispersion, and insurer participation across Colorado's individual insurance markets under the income assignment described previously. The exhibits pool March 2017–2019 data, grouping counties into thirds within each year by their pre-expansion potential individual-market population, including insured and uninsured consumers and the newly eligible group. These patterns suggest that evaluating the consequences of the eligibility expansion requires accounting for the market's susidy-setting mechanism and insurers' incentives to continue serving different markets.

The expansion would extend public eligibility to a substantial share of insurers' existing enrollees. Panel A of Table 1 shows that consumers with incomes between 138 and 199 percent FPL account for 32.9 percent of recorded private enrollment in the smallest markets, 29.9 percent in middle-sized markets, and 24.8 percent in the largest markets. The newly eligible population is disproportionately represented among current private enrollees, as their share among enrollees exceeds their share of potential consumers in each group. These shares suggest considerable scope

for expansion to reduce private enrollment, particularly in smaller markets.

Changes in private-market demand may affect the premiums insurers charge, but the implications for consumers who remain depend on what they pay after subsidies. The comparison across market sizes in Panel B of Table 1 illustrates why this distinction matters. In the smallest markets, where the newly eligible group accounts for the largest share of existing enrollment, gross premiums among consumers outside that group average $831 per month, compared with $575 in the largest markets. After subsidies, their payments average $364 and $355, respectively. More than half of these enrollees receive a subsidy in every size group. As a result, subsidies substantially narrow the difference in average premium payments between smaller and larger markets.

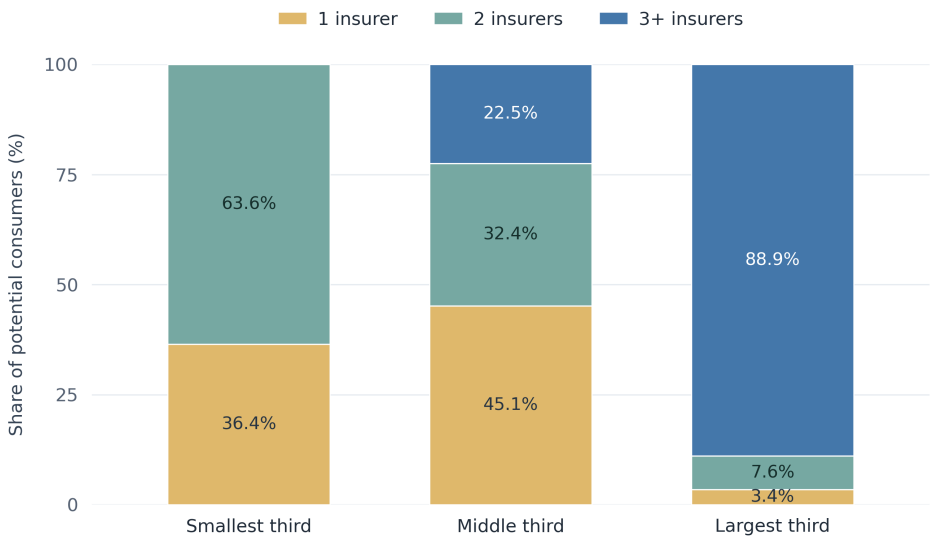
The consequences for consumers also depend on whether insurers continue to offer coverage. Figure 1 shows that all potential consumers outside the newly eligible group in the smallest markets live in counties with one or two insurers, whereas 88.9 percent in the largest markets live in counties with at least three. This suggests different implications of reduced insurer participation across markets: losing one insurer in the smallest markets would leave either a monopoly or even no insurer, while most consumers in the largest markets would retain multiple alternatives. Together with the substantial enrollment exposure documented above, these differences motivate allowing insurers' participation decisions to respond to changes in private-market demand.

Table 1: Exposure to Medicaid expansion and premium subsidies

	Smallest third	Middle third	Largest third
Panel A. Expansion population			
Share of potential consumers (%)	27.17	24.29	20.80
Share of private enrollment (%)	32.94	29.86	24.82
Panel B. Non-expansion population			
Gross premium ($/enrollee/month)	831.42	872.68	574.80
Premium subsidy ($/enrollee/month)	467.84	493.05	219.47
Net premium ($/enrollee/month)	363.58	379.63	355.34
Enrollees receiving a subsidy (%)	66.25	66.51	52.78
Enrollment rate (%)	43.02	41.65	47.63

Notes: The table pools March 2017–2019 data. Within each year, counties are grouped into thirds by pre-expansion potential population. Panel A reports the newly eligible group's shares of original population and recorded enrollment; Panel B describes consumers outside that income group. Income groups use assigned probabilities. Premiums are age-rated and enrollment-weighted. Subsidies are constructed from each year's benchmark and contribution rules, including zeros. Enrollment rates are pooled across markets and years.

Figure 1: Observed insurer participation by market size



Notes: The figure pools observed March 2017–2019 participation, defining an insurer as active with at least five enrollees in a county-year. Within each year, counties are grouped into thirds by pre-expansion potential population. Bars show the share of remaining potential consumer-years, including uninsured consumers, in counties with one, two, or at least three insurers.

4 Model and Estimation

In this section, I develop a model to evaluate the welfare consequences of Medicaid expansion. The model proceeds in three stages:

1. Insurers decide whether to enter a county after drawing **private** fixed cost shocks.
2. Insurers who enter set premiums à la Bertrand-Nash subject to regulatory constraint on markups.
3. Consumers choose among the available insurance plans

Below I discuss each stage of the model in reverse order.

4.1 Demand

Product space Insurance plans are aggregated to the firm-metal level, and premiums are the average premium across plans within each firm-metal cell. As a result, within firm-metal level, the variation in premiums only comes through variation in income-based subsidies. Each insurer

offers exactly three plans, one in each metal tier: a bronze plan with actuarial value 60%, a silver plan with actuarial value 70%, and a gold plan with actuarial value 80%. Consumers with income below 250% FPL qualify for cost-sharing reduction according to a statutory schedule, so there is also variation in actuarial values for the same silver plan offered by the same insurer. All plans offered by the same firm share the same network.

Consumer utility Conditional on being in income bin b, individual i in market m gets utility from plan j from payer $f(j)$

$$u_{ijm}^{(b)} = \alpha_i \, \text{NetPremium}_{i,j,m,t}^{(b)} + \beta_i^{AV} \, \text{ActuarialValue}_{j,m,t}^{(b)} + \beta_i^{N} \, \text{NetworkValue}_{i,f(j),m,t} + \delta_{c(i),f(j),m,t} + \epsilon_{ijmt} \tag{1}$$

where $\text{NetPremium}_{i,j,m}^{(b)}$ is the net premium a consumer pays for plan j after accounting for income-specific subsidy. [6] $\text{ActuarialValue}_{j}^{(b)}$ is the actuarial value of plan that summarizes how much a consumer expects to pay out-of-pocket when they enroll. This term is also income specific because individuals with income below 250% FPL qualify for cost-sharing reductions when they enroll in any Silver-tier plan. $\text{NetworkValue}_{i,f}$ is the value of the network of insurer f to person i given by

$$\text{NetworkValue}_{i,f} = \sum_{d} \gamma_i^d E[\max_{h} \hat{u}_{ifdh}]$$

where γ_i^d measures the frequency of diagnosis d by dividing the number of episodes with the enrollee size of each age bin $\times$ sex cell, and $\hat{u}_{ijdh}$ is the predicted utility person i places on a hospital h, conditional receiving the diagnosis d and its in-network status with insurer f. The value of $\hat{u}_{ijdh}$ is based on a separately estimated hospital choice model that closely follows Ho and Lee (2017).[7] Appendix B.1 explains the modeling and estimation in more detail.

The utility for the plan characteristics vary by c_i, a health status indicator that takes the value of 1 if an individual has diagnoses in at least two of five chronic condition domains: hypertension, diabetes, cardiovascular disease or stroke, chronic pulmonary disease including asthma, and cancer. The fixed effects $\delta_{c(i),f(j),m}$ capture other (potentially health-specific) plan characteristics

[6] $\text{NetPremium}_{i,j,m}^{(b)} = \max\{0, \text{GrossPremium}_{j,m} - \text{Subsidy}_i(b)\}$, where $\text{Subsidy}_i(b)$ is determined according to a statutory income schedule that decreases with income.

[7] While the claims data for physicians is too sparse to construct a measure of physician network, it is likely highly correlated with the strength of local hospital network.

unobservable to the econometrician.

Drivers of selection In this model, selection is driven by several forces. First, as prices rise, plans disproportionately attract less healthy individuals who are less sensitive to prices. Since an increase in the gross price passes through differently to consumers depending on income-based subsidies, gross price changes also shift the income composition of enrollees through the subsidy mechanism. Second, plans with higher actuarial value attract less healthy individuals. Lastly, plans with different network values – which can be driven by the local network breadth as well as how local demographics interact with hospital characteristics – also induce sorting of different demographic groups across plans.

Estimation The parameters are estimated with constrained maximum likelihood. At each $\theta = \{\alpha, \beta^{AV}, \beta^{N}\}$, I find $\delta_{f(j),m,\text{c(i)}}$ that matches each plan's observed shares. Because true income is not observed at the individual level, the likelihood integrates choice probabilities conditional on income over the distribution of income bins.

Because the net premium for the same plan varies exogenously according to fixed age and income subsidy schedules, the premium coefficients are identified from these exogenous within-plan variation. The actuarial value coefficients are identified because insurers are required to offer the same set of metal tiers, and because the actuarial value within silver tier plans vary by statutorily determined income brackets. The network value coefficients are identified from within-plan variation in network utility across demographic groups and geographical areas.

Results Table 2 reports the estimates for the demand parameters in equation 1. The estimates have the expected signs and indicate substantial heterogeneity by health status. For individuals with fewer than two chronic conditions, a $100 increase in monthly net premium reduces utility by 1.213; for individuals with multiple chronic conditions, the total coefficient is -0.905, implying that their demand remains downward sloping but is about 25.4 percent less price sensitive. A 10-percentage-point increase in actuarial value raises utility by 0.500 for the baseline group and 0.838 for the multiple-condition group, corresponding to implied monthly willingness to pay of approximately $41 and $93, respectively. Network value is also more important for individuals with multiple conditions: their total coefficient is approximately 4.8 times the baseline coefficient. A

one-standard-deviation increase in the network-value index corresponds to implied monthly willingness to pay of approximately $224 for the baseline group and $1,438 for the multiple-condition group.

Table 2: Main Demand Estimates

Variable	Estimate	Std. Err.
Net premium ($100)	-1.213	0.012
× Multiple chronic conditions (0/1)	0.308	0.045
Actuarial value	5.003	0.094
× Multiple chronic conditions (0/1)	3.380	0.479
Network value	9.885	0.266
× Multiple chronic conditions (0/1)	37.482	2.094

Note: The table reports the maximum likelihood estimates from the demand specification in eq. 1. The estimation sample is a random subsample of insured enrollee-year records pooled across 2017–2019 augmented with simulated uninsured individuals. The final estimation sample contains 59,611 person-year observations. Multiple chronic conditions (0/1) is an indicator equal to one when an individual has diagnoses in at least two of five chronic condition domains: hypertension, diabetes, cardiovascular disease or stroke, chronic pulmonary disease including asthma, and cancer.

4.2 Supply Stage 2: Firm Pricing Problem and Marginal Cost Specification

In this stage, insurers set premiums via a differentiated Bertrand pricing game, having observed each firm's entry decision. Recall that each insurer sells three plans, one in each metal tier, as required by regulation, and that networks are taken as given. [8]

The variable profit for each plan $j \in \{B,S,G\}$ is the total premium revenue net of the total cost of each plan summed over all consumer types k present in market m:

$$\pi_{jfm}(p_m) = \sum_{k \in m} p_{kjm} D_{kjm}(p_m) - \sum_{k \in m} MC_{kjm} D_{kjm}(p_m)$$

Firms choose prices to maximize their profit across plans subject to the Medical Loss Ratio (MLR) constraint, which requires that a firm's premium revenue to not exceed 1.25 times of the claims cost.[9] Consumer types in each market are characterized by age, sex, multiple chronic con-

[8] Insurers typically negotiate networks across commmercial market segments. Because the individual market is a relatively small segment for most insurers, pricing decisions in this market is unlikely to affect network decisions across segments.

[9] Because the observed price ratio is approximately fixed across metal tiers for the same firm and federal rules prohibit plan-specific adjustments that reflect risk selection across plans, I model firms as choosing the price of the bronze price while setting the price of the silver and gold prices according to observed multipliers between plans.

ditions indicator, income bin, on-exchange indicator, risk bin (more details below), and network value.

The type-specific marginal cost for each plan has two components and is given by $MC_{kjfm} = HC_{kjfm} + c_{fm}$. HC_{kjfm} is the type-plan-specific expected claims cost (more details below). c_{fm} is the residual marginal cost that approximates administrative costs that do not vary by health care utilization, as well as expectational errors not captured by HC_{kjfm}.

Expected claims costs The modeling of the expected claims cost term HC_{tjfm} serves two purposes. Economically, it captures changes in firms' expected cost as individuals with different expected costs sort between plans. Additionally, it needs to preserve rich cost heterogeneity while keeping the type space parsimonious for the heavy downstream computation.

For these reasons, I take a two-step approach to model expected claims cost. The first step uses rich lagged claims information to predict each individual's expected cost. I train a flexible claims prediction model using all medical and pharmacy spending in the previous two years. This step produces an individual-level predicted per-member per-month cost, which is then discretized into a small number of risk bins. In the second step, I train a type-plan level cost model using the type variables and the risk bins to predict expected claims costs by firm and metal tier. This gives HC_{tjfm} enough variation to reflect sorting by expected cost across plans, while keeping the type space manageable. Appendix C explains the model training and selection in more detail.

Residual marginal costs In standard Bertrand pricing games, marginal costs can be directly inverted from the firms' pricing first-order conditions. However, in my constrained pricing game, one cannot infer from the data whether a firm is slack or binding at the observed prices. Therefore, I estimate a slack-MLR candidate that sets the derivative of unconstrained profit with respect to price equal to zero, and a binding-MLR candidate that makes claims plus residual costs equal to 80 percent of premium revenue, then check whether the resulting estimates are consistent with the assumption under which it was estimated. Of the 434 year-county-firm cells, all but six cells produce one unique admissible $\hat{c}_{fm}$.

To get residual marginal costs for all potential firm-market cells, I fit a projection model relating the 428 uniquely inverted costs to observable firm-market characteristics. The regressors

include the firm's lagged presence in the focal county, the share of other counties in the same rating area in which it was active in the previous year, summaries of expected claims and demand utility, and separate firm, county, and year fixed effects. The residuals from this projection form the empirical distribution of ω_{fm}, the marginal cost shock realized after entry but before pricing. More details are in Appendix D.

4.3 Supply Stage 1: Entry

Because individual market insurers file entry decisions at a given time each year, and the fixed cost of selling plans is small relative to the marginal cost in this setting, I model insurers as playing a static entry game. In the first stage, insurers make static entry and exit decisions based on their expected profits. The set of potential entrants is the six insurers that operate in Colorado during my sample period. At the beginning of each period, firms observe private fixed cost shocks and the distribution of fixed cost shocks of the other firms, and enter if expected profits are positive. Entered firms then set prices after their marginal cost shocks are realized. Hence, selection into entry is driven by fixed cost realizations and expected profit over the distribution of marginal cost shocks.[10]

Let $Y_{mt} \in \{0,1\}^{|F|}$ be the market-specific entry vector. The total profit of firm f entering county m with entry vector Y_{mt}, private fixed-cost shock ϵ_{fmt}, and marginal-cost shock ω_{mt} is given by

$$\Pi_{fmt}(Y_{mt}, \epsilon_{fmt}, \omega_{mt}) = V_{fmt}(Y_{mt}, \omega_{mt}) - FC_{fmt}(\epsilon_{fmt}).$$

Let M_{mt} be potential market size, and let $S_{fm,t-1}$ be firm f's enrollment share in county m in the previous year. The fixed cost of entry is parameterized as

$$FC_{fmt}(\epsilon_{fmt}) = M_{mt}^{\rho} \exp\{\mu + \alpha S_{fm,t-1} + \delta \mathbf{1}\{S_{fm,t-1} > 0\} + \lambda_t + \sigma\epsilon_{fmt}\}, \qquad \epsilon_{fmt} \sim N(0,1),$$

where λ_t is a year fixed effect (2017 is the reference year) and ϵ_{fmt} is a private standard-normal shock. Thus, fixed costs are lognormal, with market size, continuous lagged enrollment share, and an indicator for any positive lagged enrollment shifting log fixed costs.

[10] Modeling fixed cost shocks as private guarantees equilibrium existence and, with extra assumptions, produces point estimates of the distribution parameters of entry fixed costs.

Let $\tilde{q}_{jmt}$ be the belief about the probability firm j will enter county m in year t. Firm f's probability of facing rival set $y_{-f,mt}$ is

$$\Pr(y_{-f,mt}|\tilde{q}_{mt}) = \prod_{j \neq f} \tilde{q}_{jmt}^{y_j}(1-\tilde{q}_{jmt})^{1-y_j}$$

Its expected variable profit is

$$\overline{V}_{fmt}(\tilde{q}_{-f,mt}) = \sum_{y_{-f,mt}} \Pr(y_{-f,mt}|\tilde{q}_{mt}) E_\omega[V_{fmt}(y_{-f,mt}, \omega_{mt})].$$

Under the belief $\tilde{q}_{-f,mt}$, firm f's ex-ante best response for positive expected variable profit is

$$q_{fmt}(\tilde{q}_{-f,mt}) = \Pr\left[\overline{V}_{fmt}(\tilde{q}_{-f,mt}) \geq FC_{fmt}(\epsilon_{fmt})\right].$$

In equilibrium, all firms best respond to the entry probabilities of the other firms, so an equilibrium of this game is characterized by a fixed point in entry probabilities.

Estimation To estimate the fixed-cost parameters, I use a two-stage procedure analogous to Pakes, Ostrovsky, and Berry (2007). Observed entry is defined as having at least five current-March enrollees. In the first stage, I estimate the entry policy functions played in the observed equilibrium. In the second stage, I combine the estimated entry probabilities with precomputed stage-game profits to construct each firm's expected variable profit from entry, and then estimate the fixed-cost parameters by fitting model-implied entry probabilities to observed entry decisions. This approach avoids repeatedly solving the entry game or imposing an equilibrium-selection rule during estimation.

To estimate the first-stage policy functions, I use a profit-label neural screen to rank invariant candidate variables, followed by penalized Logit selection of optional predictors. Log potential market size, lagged enrollment share, an indicator for positive lagged share, and year indicators are mandatory; the final policy function is an unpenalized Logit on these controls and 13 selected optional variables. The optional variables chiefly describe rivals' demand attractiveness, health-cost exposure, lagged presence, and substitution overlap, together with a focal-firm cost–utility measure. Rival-entry indicators enter the neural screen as context controls but are not selectable

policy predictors.

I estimate the main second-stage parameters by maximum likelihood. As a robustness estimator, I use nine GMM moments that set entry-probability residuals orthogonal to a constant, log expected variable profit, lagged enrollment share, an indicator for positive lagged enrollment, log potential market size, the 2018 and 2019 indicators, equal-weight mean stage-game profit, and average profit loss from an additional rival. The continuous and payoff instruments are standardized for numerical conditioning, and the moments receive identity weighting. Appendix E explains both stages in more detail. Appendix F explains the algorithm I use to solve the stage-game profits.

Results Table 3 reports the maximum-likelihood and GMM estimates. The maximum-likelihood estimate of ρ is 0.284, implying that median fixed cost rises less than proportionally with raw potential market size, holding other covariates fixed. The estimate of δ is -2.055: any positive lagged enrollment is associated with a reduction in median fixed cost, in addition to the continuous share effect. Conditional on positive lagged enrollment, a ten-percentage-point increase in share multiplies median fixed cost by $\exp(-0.1 \times 0.697) \approx 0.933$. The positive year effects indicate higher fitted median costs in 2018 and 2019 than in 2017, conditional on the other covariates. The GMM point estimates are broadly similar.

4.4 Model Fit

Figure 2 compares observed entry frequencies with mean MLE-predicted entry probabilities across five dimensions. Panel (a) shows that the model matches the aggregate entry rate closely: 36.1% of firm–county–year cells have observed entry, versus a predicted mean of 35.8%. Panel (b) asks whether entry rises appropriately with expected variable profit. Both series increase steeply across profit quartiles, from 4.7% observed versus 6.4% predicted in the bottom quartile to 76.4% versus 77.1% in the top quartile.

Panel (c) sorts cells by fitted expected fixed cost. Entry falls from 71.0% observed and 70.1% predicted in the lowest-cost quartile to 4.4% and 8.9% in the highest-cost quartile. The largest discrepancy here is in the third quartile, which the model underpredicts by 6.3 percentage points. Panel (d) groups county-years by potential market size. The model tracks the high entry rate in

Table 3: Lognormal Fixed-Cost Estimates

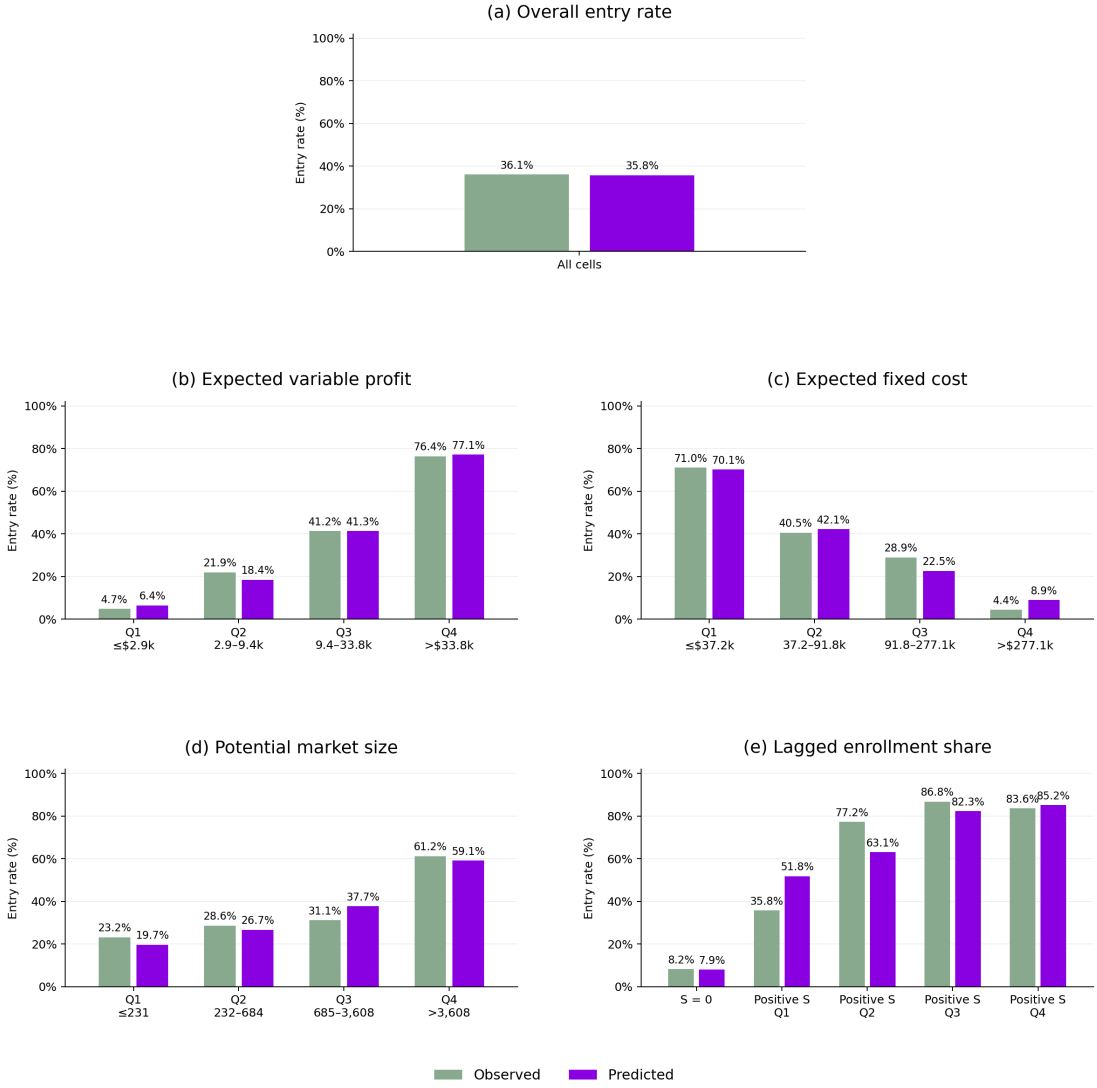
	Probit MLE		GMM	
Parameter	Estimate	SE	Estimate	SE
Log-cost constant μ	8.607	0.741	8.635	0.832
Lagged enrollment share α	-0.697	0.430	-0.550	0.441
Positive lagged share δ	-2.055	0.305	-2.129	0.351
Market-size elasticity ρ	0.284	0.090	0.299	0.090
Log-cost shock SD σ	1.542	0.231	1.470	0.234
Year effects (2017 reference)				
2018	0.995	0.202	0.860	0.202
2019	1.172	0.191	0.981	0.206

Notes: The sample contains 1,098 firm–county–year observations: six potential insurers in 61 counties over 2017–2019 (183 county-years); two modeled counties are excluded in all three years. Entry is defined as at least five current-March enrollees, yielding 396 observed entries. Potential market size is the unnormalized count of people. Standard errors are the sample standard deviations from 1,000 bootstrap replications that resample complete county panels statewide. Each replication refits first-stage preprocessing and the unpenalized Logit, reintegrates expected variable profit, and re-estimates MLE and GMM. The selected first-stage feature set and pricing-stage payoff schedule are held fixed.

the largest markets (61.2% observed versus 59.1% predicted), but overpredicts entry in the third size quartile by 6.6 percentage points (37.7% versus 31.1%).

Panel (e) examines persistence across lagged-enrollment-share groups. Predicted and observed rates are nearly identical for firms with zero lagged share (7.9% versus 8.2%) and in the highest positive-share quartile. Fit is weaker among firms with small but positive lagged shares: the model overpredicts the first positive-share quartile by 16.0 percentage points (51.8% versus 35.8%) and underpredicts the second by 14.1 points (63.1% versus 77.2%).

Figure 2: In-sample entry-model fit: year-effects specification, five-enrollee entry cutoff



Notes: Bars show the share of cells with at least five current-March enrollees and mean MLE-predicted entry probabilities for the same 1,098 cells. Expected-variable-profit quartiles use profits reintegrated with the updated first-stage policy probabilities. Expected fixed cost in panel (c) is the unconditional mean of the fitted lognormal distribution, not cost conditional on entry. Market-size quartiles are formed across county-years, with all six potential firms in a county-year assigned to the same quartile. Panel (e) separates zero lagged share from quartiles of positive lagged share.

5 Counterfactual Simulation

I use the estimated demand, pricing, and entry models to simulate an extension of Medicaid eligibility to consumers with incomes between 138 and 199 percent of the federal poverty level, which removes this group from the private individual market. I first describe changes in market outcomes, then examine how the resulting shifts in market structure shape consumer welfare and government spending across markets of different sizes.

The expansion removes a substantial share of the private market's consumers and changes the composition of the private market. Table 4 shows that the expansion group accounts for 21.3 percent of all potential individual market consumers and 25.9 percent of private enrollees and reduces insurer spending by 22.9 dollars per month per enrollee. This is reflected in gross premiums, which fall by 10.5 dollars per month per enrollee. However, lower subsidies offset the decline in gross premiums when holding entry fixed at baseline. Premium subsidies fall by $13.05 on average, leaving net premiums $2.53 higher.[11] The expansion also leads to large reductions in insurer profits and insurer participation. The average firm sees a reduction of 31.5 percent in expected variable profit, and the average market loses 13.7% of expected insurers.

[11] In Appendix TK, I further illustrate the subsidy offset channel and decompose the effects of repricing from those of benchmark adjustment.

Table 4: Population exposure and private-market responses

Outcome	All markets
Departing population (%)	21.28
Departing enrollment (%)	25.85
Insurer cost ($ per enrollee-month)	−22.88
Gross premium ($ per enrollee-month)	−10.53
Premium subsidy ($ per enrollee-month)	−13.05
Net premium ($ per enrollee-month)	+2.53
Variable profit (% per firm-market)	−31.48
Expected insurer count (per market)	−0.28

Notes: Departing population and enrollment are shares of baseline private market population and enrollment, respectively. Insurer cost is the expected cost incurred to insurers excluding patient payments. Expected insurer cost, subsidy, and premium outcomes integrate over cost draws and the baseline entrant-set probabilities. Variable profit averages each firm's percentage change equally across potential firm-counties. Expected insurer counts sum firm entry probabilities within each market and average across markets.

The decline in insurer participation matters because different market structures imply different enrollment, consumer surplus, and subsidy spending. Table 5 evaluates these outcomes at counterfactual premiums and subsidy benchmarks. Compared with configurations containing multiple insurers, monopoly configurations have a 12.1 percentage-point lower enrollment rate and $35.92 less consumer surplus per person per month. Net premiums differ by only $3.52 per enrollee, while non-price inclusive value—the value of available coverage options when net premiums are set to zero—is $77.80 lower under monopoly. Lower enrollment under monopoly also reduces government subsidy spending. Although subsidies per enrollee are higher under monopoly, spending per person falls from $211.81 with multiple insurers to $165.61 with one insurer due to reduced enrollment.

Table 5: Coverage, welfare, and subsidy spending across participation configurations

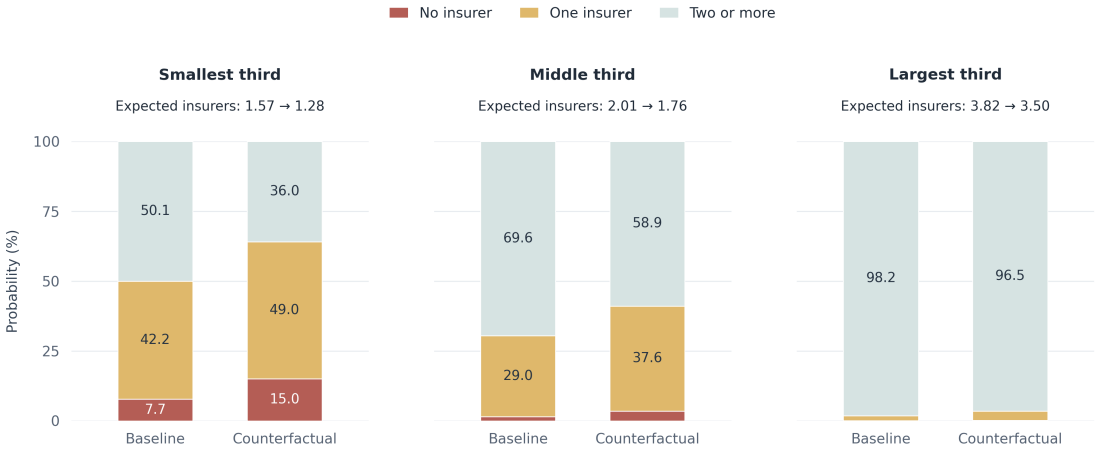
Outcome	One insurer	Two or more insurers
Enrollment rate (%)	32.74	44.81
Net premium ($ per enrollee per month)	149.99	146.46
Value of insurance options ($ per person per month)	261.18	338.98
Consumer surplus ($ per person per month)	56.60	92.52
Government subsidy ($ per person per month)	165.61	211.81

Notes: This table compares market outcomes under monopoly and multiple-insurer configurations. All outcomes use counterfactual prices. Value of insurance options measures the dollar value of the available coverage options if all plans had zero net premiums. Government spending includes premium tax credits capped at gross premiums and excludes Medicaid spending. For each market and entrant set, outcomes are averaged over cost draws and baseline entrant set probabilities normalized within each market and insurer-count category.

Participation responses that shift markets toward configurations with lower enrollment and consumer surplus are most pronounced in smaller markets. Figure 3 divides markets into thirds ranked by the size of the potential private market population pre-expansion and shows that participation response shifts smaller markets more strongly toward the configurations associated with lower coverage, consumer surplus, and subsidy spending. Expected participation falls by similar amounts across market-size groups, and profit declines are also broadly similar across the three groups, at 33.5, 29.2, and 31.8 percent. However, because smaller markets have fewer insurers at baseline, participation declines in smaller markets translate into larger increases in the probabilities of monopoly and no coverage. In the smallest markets, the probability of no insurer nearly doubles from 7.7 to 15.0 percent, and monopoly probability rises from 42.2 to 49.0 percent. In middle-sized markets, monopoly probability also rises from 29.0 to 37.6 percent. The largest markets instead retain 3.50 expected insurers and a 96.5 percent probability of multiple insurers.

Consistent with the findings on muted net price movements and entry effects, Figure 4 reveals a stark difference in welfare predictions under baseline and counterfactual entry probabilities as well as distinct welfare effects across markets in different size groups. At baseline entry probabilities, consumer surplus rises by $0.16 per person overall, with small gains in every group.

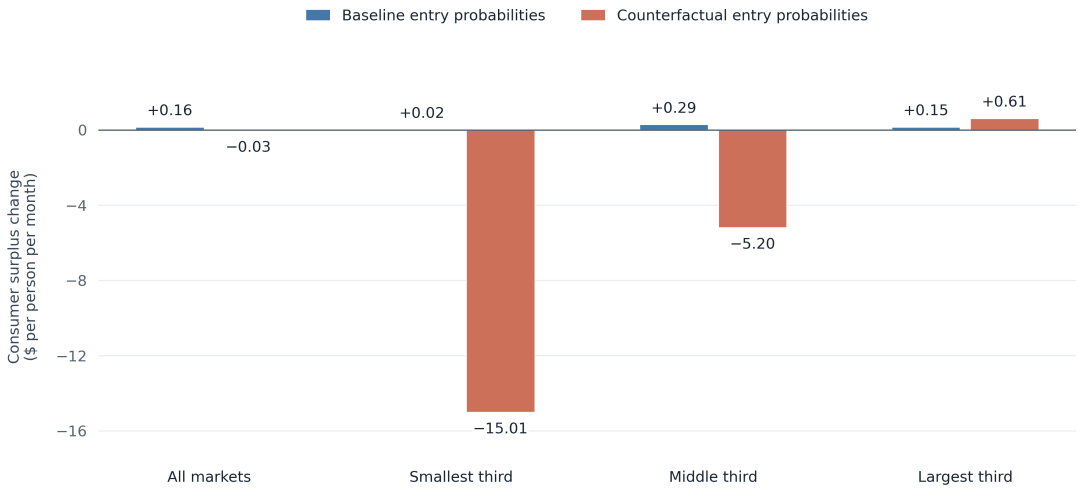
Figure 3: Changes in insurer participation by market size



Notes: Markets are grouped by the original pre-expansion potential private market population. Expected insurer counts sum the potential firms' entry probabilities within each county weighted by the remaining potential population. Each pair of bars compares baseline and counterfactual probabilities of zero, one, and two or more insurers. Within each market, entrant set probabilities are summed to obtain probabilities of insurer counts, then averaged within each group weighted by the remaining potential population.

Allowing entry to adjust produces losses of $15.01 in the smallest markets and $5.20 in the middle group, alongside a gain of $0.61 in the largest.

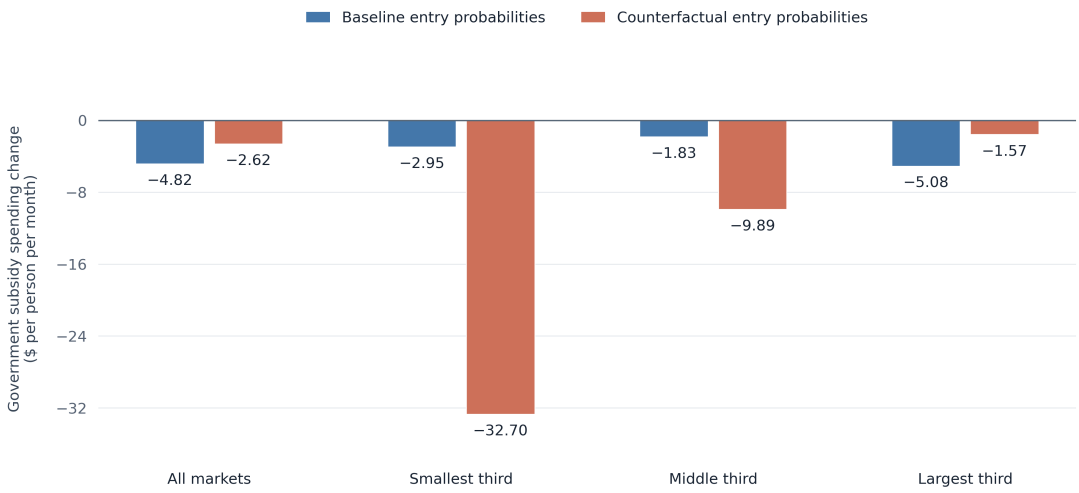
Figure 4: Consumer-surplus changes overall and by market size



Notes: Bars report changes from baseline consumer surplus in monthly dollars per remaining potential person. Both counterfactual series use counterfactual premiums and subsidy benchmarks; the left bar retains baseline entry probabilities and the right bar uses counterfactual entry probabilities. Baseline consumer surplus uses baseline prices and entry probabilities for the same remaining population. Outcomes are averaged over cost draws and then over entrant sets with the indicated probabilities. Group changes equal summed consumer-surplus changes divided by the group's remaining population.

The loss of coverage and options that reduces consumer welfare in smaller markets also lowers subsidy spending. Figure 5 shows that entry adjustment deepens spending reductions in smaller markets while moderating the statewide decline. At baseline entry probabilities, lower subsidies reduce spending in every group, by $4.82 per person overall. Allowing entry to adjust increases the reductions in the smallest and middle-sized markets to $32.70 and $9.89, respectively. In the largest markets, enrollment remains slightly above baseline and spending falls by $1.57.

Figure 5: Government premium subsidy spending changes overall and by market size



Notes: The left bar shows the spending change when premiums and subsidy benchmarks adjust but entry probabilities remain at baseline. The right bar additionally allows entry probabilities to adjust. Outcomes are averaged over cost draws and then over entrant sets with the indicated probabilities. Group changes equal summed subsidy-spending changes divided by the group's remaining population.

6 Conclusion

This paper studies the consequences of expanding public insurance for the private individual market. The analysis highlights insurer participation as a central determinant of the policy's welfare effects, reflecting two features of the market. First, price-linked subsidies dampen the effect of premium changes on consumers, limiting the welfare consequences of insurers' pricing responses. Second, consumer welfare is substantially lower under monopoly or no coverage than when multiple insurers compete. The expansion sharply increases the likelihood of these outcomes, making the loss of private insurance options consequential for consumers who remain outside the public program.

Although this analysis treats income as fixed, households' incomes fluctuate, and broader Medicaid eligibility can reduce the need to switch between Medicaid and other forms of coverage as earnings change. Preserving continuity of coverage provides an additional rationale for expansion, motivating further research on the value of Medicaid under income uncertainty. Expansion may also affect the continued availability of hospital care. By covering previously uninsured patients, it can reduce uncompensated care, strengthen hospitals' finances, and reduce the risk of closure (Dranove, Garthwaite, and Ody, 2016; Lindrooth et al., 2018). An expansion could therefore discourage private insurer participation while helping sustain hospitals, creating a potential trade-off between insurance availability and provider availability. Hospital availability remained stable in Colorado during my sample period, but studying settings in which Medicaid plays a larger role in hospital viability would allow future research to assess how the benefits of preserving access to hospital care compare with the losses from reduced private insurance choice.

References

A Details on Income Distribution Assignment

This appendix describes how I combine administrative enrollment records with external data to assign income probabilities. Each insured member-year receives a probability distribution over six income categories: 0–137, 138–199, 200–249, 250–299, 300–399, and 400 percent or more of the federal poverty level (FPL). Simulated uninsured individuals are assigned a joint probability distribution over the six income categories and three morbidity categories.

For each insured enrollee, the assignment proceeds in four steps. I first establish a statewide income distribution specific to the enrollee's exchange status and year. I then adjust this distribution using the income distribution in the enrollee's county and age group. Next, I use MEPS to adjust it according to the relationship between income and the enrollee's sex and observed morbidity. Finally, I calibrate the resulting individual probabilities so that, in aggregate, they reproduce the selected statewide income distribution for that exchange-status-year cell. The income assignment for the simulated uninsured population follows a similar logic. However, since morbidity is not observed for simulated uninsured individuals, each person receives a joint probability distribution over six income categories and three morbidity categories. The size of the uninsured population is derived using the ACS county-by-age ratio of uninsured adults to adults with direct-purchase-only coverage (i.e., the number of on- and off-exchange enrollees in the CO APCD).

Statewide income distribution For on-exchange enrollees, I use the reported FPL distribution from the Connect For Health (Colorado's state-run ACA exchange) enrollment reports. [12] For off-exchange enrollees, I use pooled 2017–2019 Colorado observations from the CPS ASEC and calculate the statewide income distribution among adults between 18 and 64 with direct-purchase, non-Marketplace coverage. For the uninsured sample, I use the yearly ACS Public Use Microdata (PUMS) to calculate the statewide income distribution among adults between 18 and 64 who report being uninsured.

[12] The distribution is calculated among Bronze, Silver, and Gold enrollment records for which FPL is reported.

ACS geographic adjustment The second step uses the ACS to construct a county-by-age adjustment. This step is applied to all three coverage groups. For each county, age group (18–24, 25–34, 35–44, 45–54, and 55–64), and income category, I calculate the category's share of the county-age population and divide it by the category's share of the statewide adult population. This ratio measures whether the income category is relatively more or less prevalent in that county-age cell than in Colorado overall. [13] The raw county-age table containing 20 zero income cells/ 11 small county-age strata are pooled with geographically adjacent counties. For each approved pool, I combine the ACS counts, calculate a common six-category income distribution, and assign that distribution to every county in the pool. These pools cover 19 county-age strata and affect approximately 0.20 percent of the statewide adult ACS population.

MEPS joint income-morbidity distribution The third step uses the MEPS to estimate how morbidity (0/1/2+ chronic conditions) varies with income. For observed insured enrollees, morbidity is calculated from diagnoses observed in claims during the 24 months preceding the March index date. For MEPS respondents, I construct the analogous measure using the available doctor-diagnosed condition indicators. I pool the 2017–2019 MEPS files but estimate the relationship separately for on-exchange, off-exchange, and uninsured respondents. I also retain separate relationships by sex. [14] For each insured enrollee and income category, I multiply the statewide income probability and ACS geographic adjustment by the MEPS probability of observing the enrollee's morbidity category, conditional on sex, exchange status, and income group. For a simulated uninsured individual, both income and morbidity are latent, so I construct a score for each of the 18 possible income-morbidity combinations by multiplying the annual statewide uninsured income probability, the ACS county-age adjustment, and the MEPS probability of the morbidity category conditional on sex and income group.

Final calibration After the preceding steps, each person has preliminary income probabilities, q_{ik}, that reflect coverage status, county, age, sex, and morbidity. However, averaging these prob-

[13] The published ACS categories do not align exactly with all six analysis categories. The ACS reports 125–149 percent FPL as one category. I divide this count equally between the 0–137 and 138–199 percent categories. Similarly, I divide the published 200–299 percent category equally between the 200–249 and 250–299 percent categories.

[14] To avoid estimating this relationship from very sparse cells, I combine the six income categories into three groups for the MEPS calculation: 0–137, 138–199, and 200 percent or more of FPL and pool over ages in the MEPS calculation because age-related income variation is supplied by the ACS county-age distribution.

abilities across the population need not reproduce the statewide income distribution. To eliminate this discrepancy, I estimate one adjustment factor, a_k, for each income category within each coverage-group-year cell, so that the final probability for each person is given by

$$p_{ik} = \frac{a_k q_{ik}}{\sum_l a_l q_{il}}$$

The factors a_k are chosen so that the population-weighted average of p_{ik} equals the external statewide target in step 1 above for every income category.

B Demand Estimation Details

B.1 Hospital choice model and estimation

Hospital-choice data The hospital-choice model uses inpatient episodes observed in the Colorado All-Payer Claims Database from 2017 through 2019. The sample is restricted to individual-market enrollees between ages 18 and 64 with a qualifying inpatient admission. [15]

Each admission is assigned to one of eight broad diagnosis groups. Starting with the MS-DRG/MDC-based classification of admissions into broad diagnosis groups, I then separately use the principal diagnosis code to identify admissions associated with cancer, allowing cancer cases that span multiple MDCs to be classified into a common cancer category. For each admission, the choice set consists of Colorado hospitals classified as short-term or critical access with valid distance information for the patient's ZIP code. Hospital-year alternatives that are never chosen in the estimation sample are removed.

For each hospital, I construct measures of service availability in five areas using the CMS Provider of Services files: cardiac care, maternity and neonatal care, mental-health and substance-use treatment, advanced imaging, and oncology. Each measure counts the relevant services available at the hospital. For example, a hospital with a catheterization laboratory, a coronary-care unit, and open-heart surgery is counted as having a cardiac measure of three.

[15] Claims belonging to overlapping inpatient stays are consolidated into episodes. Episodes associated with more than one payer are excluded.

Hospital-choice model Conditional on an inpatient admission and diagnosis group d, individual i, enrolled with payer j, chooses hospital h in year t. Latent utility is

$$u_{ijdht} = \delta_{ht} + \beta^D_{r(i)} \text{Dist}_{ih} + \beta^N_d \text{InNetwork}_{jht} + \sum_{s \in \mathcal{S}} \beta^S_s \mathbf{1}\{d = d(s)\} \text{Service}_{hs} + \varepsilon_{ijdht}, \tag{2}$$

where δ_{ht} is a hospital-year fixed effect, $r(i)$ is the hospital-referral region (HRR) containing the individual's ZIP code, and Dist_{ih} is distance in kilometers. The network coefficient varies across the eight diagnosis groups. The set $\mathcal{S}$ contains cardiac, maternity, mental-health, imaging, and oncology services; each service measure enters only for its corresponding diagnosis group. ε_{ijdht} is the i.i.d. logit error. I estimate the model by maximum likelihood.

Table 6 reports the resulting estimates. As expected, distance reduces hospital utility in every referral region, with an additional kilometer lowering utility by between 0.022 and 0.083 units. In-network status raises utility each of the diagnosis groups, although the estimate for mental and behavioral diagnosis is small and imprecise. The positive diagnosis-service interactions indicate that patients place greater value on hospitals offering services relevant to their admission.

B.2 Insurer-market-morbidity fixed effects prediction

The demand inversion identifies an insurer-market-morbidity fixed effect when an insurer has positive observed enrollment in that market and morbidity group. Since counterfactual entry and exit exercises require a value for every potential insurer–market–morbidity combination, including combinations not observed in the estimation data, I use the identified fixed effects to estimate a projection equation and use its fitted values for these combinations.

Specifically, I regress the fixed effects recovered from the demand inversion on year, insurer, and county-group fixed effects; an indicator for having two or more chronic-condition domains; interactions of that indicator with year and insurer; county population; the share under age 65 with subsidy-eligible income; and an indicator for whether the same insurer participated in the individual market in that county in the preceding year.

Table 7 reports the results from this regression. The model explains 91.4 percent of the variation in the identified fixed effects. Lagged insurer presence is strongly positively associated with an insurer's residual mean utility in a county, consistent with substantial persistence in unob-

Table 6: Hospital-choice estimates used to construct network value

Variable	Estimate	Standard error
Distance by hospital-referral region		
Distance × Boulder	-0.083	0.004
Distance × Colorado Springs	-0.040	0.002
Distance × Denver	-0.066	0.002
Distance × Fort Collins	-0.052	0.003
Distance × Grand Junction	-0.030	0.001
Distance × Greeley	-0.066	0.003
Distance × Other	-0.022	0.001
In-network status by diagnosis group		
In network × Circulatory and respiratory	1.659	0.153
In network × Digestive	2.211	0.156
In network × Musculoskeletal	1.954	0.170
In network × Neurological and sensory	1.137	0.209
In network × Reproductive, maternal, and neonatal	2.147	0.131
In network × Mental and behavioral	0.121	0.263
In network × Cancer	2.957	0.382
In network × Other	1.537	0.157
Diagnosis-specific hospital services		
Circulatory and respiratory × cardiac services	0.182	0.071
Reproductive, maternal, and neonatal × maternity services	0.101	0.041
Mental and behavioral × mental-health services	0.936	0.140
Cancer × imaging services	0.471	0.145
Cancer × oncology services	0.377	0.192
Hospital-year FE		Yes
Number of episodes		15,021

Notes: This table reports the estimates of the hospital demand model in equation (2). The estimation sample consists of inpatient episodes of Colorado individual market enrollees from 2017–2019.

served insurer-market attributes. The remaining coefficients describe conditional differences in residual utility across years, insurers, morbidity groups, and market characteristics. Because the dependent variable is the demand fixed effect remaining after controlling for premium, actuarial value, and network value, these estimates should not be interpreted as structural preference parameters.

Table 7: Projection of insurer–market–morbidity fixed effects

Variable	Estimate	Standard error
County population under 65 (100,000)	-0.425	0.054
Share under 65 with subsidy-eligible income (10 pp)	-0.564	0.176
Insurer present in county in preceding year	1.491	0.388
Year FE		Yes
Insurer FE		Yes
County-morbidity group FE		Yes
Morbidity-group FE		Yes
Year × morbidity-group FE		Yes
Insurer × morbidity-group FE		Yes
Observations		575
R^2		0.914
Adjusted R^2		0.909

Notes: Table reports the results from the fixed effects regression. The dependent variable is the insurer–county–year–morbidity fixed effect identified by the demand inversion.

C Details on Expected Claims Cost Prediction

This appendix describes the construction of the expected claims cost object used in the supply model. I use a two-step procedure so that the predicted cost object both summarizes rich heterogeneity in health risk and remains tractable enough to enter the equilibrium computation. First, I predict individual expected claims costs using lagged claims history. Second, I compress those predictions into a small number of risk types and estimate expected costs at the type-plan level used by the pricing and entry model.

The first-stage prediction model is selected on person-level fit. Conditional on that model, the second-stage specification is selected jointly over the risk-bin definition and the type-plan-cell cost model. Section C.2 describes the candidate risk bins, the candidate type-plan-cell models, the selection criteria, and the selected downstream specification.

C.1 Person-level prediction

The first step summarizes each individual's observed health history with a payer-invariant prediction of future claims cost. The purpose of this prediction is to measure underlying health risk before assigning individuals to the smaller set of risk types used in the equilibrium model.

The unit of observation is a person-year. The outcome is annual claims cost per member month, constructed from plan-covered plus member-liability claims amounts and aggregated across the individual's modeled coverage records during the year. Observations are weighted by enrolled member months. To limit the influence of exceptionally large realized claims, claims PMPM is winsorized at the year-specific 99th percentile before the model is fitted.

The predictors summarize the individual's health and utilization history in the previous two years. They include total, medical, pharmacy, inpatient, outpatient, professional, and emergency-room spending; claim and service counts; diagnosis, procedure, and drug information; clinical morbidity measures; prior high-cost status; and changes in spending and utilization between the two lagged years. The model also includes demographic, geographic, enrollment-channel, and morbidity characteristics.

I estimate the person-level score using a histogram gradient-boosted regression with a Poisson loss. Gradient boosting flexibly captures nonlinear relationships and interactions among the claims-history variables, while the Poisson loss is suited to a nonnegative and highly skewed cost outcome.

The person-level prediction enters the second step in two ways. First, it is used to assign individuals with prior claims exposure to candidate discrete risk bins. Second, the average predicted person-level risk within a type-plan cell is included as a predictor in the type-plan-cell cost model. Insurer, plan, and metal-tier differences enter only in that second-stage model, which converts the payer-invariant health-risk summary into the plan-specific expected claims costs used by the equilibrium model.

Table 8 reports the temporal out-of-sample performance of the person-level health-cost prediction model. The model explains approximately one-quarter of the variation in the winsorized PMPM outcome, with weighted R^2 values of 0.258 in 2018 and 0.253 in 2019. Mean predictions are also reasonably well calibrated to mean realized spending. The median absolute error is $197.50

and $207.24, respectively. The larger mean average errors—$421 in 2018 and $433 in 2019—reflect the right-skewed distribution of healthcare spending and prediction errors.

Table 8: Out-of-sample validation of the person-level health-cost prediction model

Statistic	2018	2019
Training years	2017	2017–2018
Person-years	125,799	164,838
Member-months	1,337,463	1,715,764
Weighted R^2	0.258	0.253
Actual mean PMPM ($)	389.73	422.01
Predicted mean PMPM ($)	400.46	403.79
Median absolute error ($ PMPM)	197.50	207.24
Mean average error ($ PMPM)	421.22	432.64

Notes: The table reports temporal out-of-sample validation using the year-specific, 99th-percentile-winsorized PMPM outcome. The 2018 model is estimated on 2017 observations, while the 2019 model is estimated on 2017–2018 observations. All performance statistics are weighted by member-months.

C.2 Second-stage type-plan cell prediction

The second stage jointly selects how the person-level risk predictions are compressed into discrete risk bins and how expected claims costs are predicted at the type-plan cell level. The selected specification converts the person-level predictions into four risk bins and combines those bins with demographic, geographic, insurer, and plan characteristics to predict expected monthly claims costs for every type-plan cell required by the equilibrium model.

The person-level prediction from stage one enters this construction both discretely and continuously. Individuals with no claims exposure in either of the previous two years are assigned to risk bin 0. Among individuals with prior exposure, the selected rule uses the within-year, member-month-weighted distribution of predicted PMPM: the bottom 80 percent are assigned to risk bin 1, the next 18 percent to risk bin 2, and the highest 2 percent to risk bin 3. The model also retains the average continuous person-level predicted PMPM within each type-plan cell. Thus, the risk bins capture broad differences in the distribution of health risk, while the continuous average preserves variation within those bins.

The estimation sample is constructed by aggregating the claims data to type-plan cells. A cell is defined by year, county and rating area, insurer, metal tier, age, sex, the indicator for having at least two chronic conditions, exchange channel, and risk bin. The outcome for each cell is realized

claims cost per member month, with the underlying person-payer-year PMPM outcome is winsorized at the year-specific 99th percentile before aggregation. The predictors include the variables defining the type-plan cell, together with the plan's network value measure and the cell mean of the person-level predicted PMPM from the first step. Insurer and metal-tier indicators allow expected claims to differ across plans, while age, sex, morbidity, geography, exchange channel, risk bin, and average predicted risk capture differences in enrollee composition.

Expected cell-level claims costs are estimated using a histogram gradient-boosted regression with a Poisson loss. The model flexibly captures nonlinearities and interactions among the cell characteristics while restricting predicted claims costs to be nonnegative. The base model is followed by a multiplicative year-insurer calibration intended to correct persistent differences between mean actual and predicted claims for particular insurers in particular years. To reduce sampling noise, each year-insurer adjustment is shrunk toward the overall calibration factor, with the amount of shrinkage decreasing as the group's member-month exposure increases.

The risk-bin definition and type-plan cell estimator are selected jointly using temporal out-of-sample validation. For each validation year, the risk-bin thresholds, and type-plan cell cost model are estimated using earlier years and evaluated in the held-out year. The candidate specifications include alternative predicted-risk thresholds, prior-cost-based bins, grouped and smoothed cell means, uncalibrated Poisson gradient boosting, and models with year-insurer or year-insurer-metal calibration. The selection procedure prioritizes average and worst-year out-of-sample R^2, followed by mean absolute error, insurer-year calibration, performance in the high-cost tail, monotonicity across risk bins, and exposure to sparsely populated cells. The selected pair is the top-2-percent risk-bin definition and the year-insurer-calibrated Poisson gradient-boosted model. After model selection, the person-level risk model and type-plan cell model are refitted using all available claims years.

Table 9 reports the out-of-sample fit of the selected model at the type-plan-cell level. The model explains 38.3% of weighted cost variation in 2018 and 42.0% in 2019. The median absolute error is approximately \$132–\$144 PMPM, while the mean absolute error is about \$325 PMPM, again reflecting the right-skewed cost distribution. Predicted mean costs are very close to the actual means. These errors are materially smaller than the person-level results, as expected because aggregation into type-plan cells averages out idiosyncratic person-level spending variation.

Table 9: Out-of-sample validation of type-plan-cell health-cost predictions

Statistic	2018	2019
Training years	2017	2017–2018
Type-plan cells	40,729	52,588
Member-months	1,337,463	1,715,764
R^2	0.383	0.420
Mean absolute error (PMPM)	\$326.84	\$324.79
Median absolute error (PMPM)	\$131.65	\$143.87
Actual mean cost (PMPM)	\$389.73	\$422.01
Predicted mean cost (PMPM)	\$423.27	\$432.98
Mean absolute insurer-year bias (PMPM)	\$33.54	\$14.66

Notes: Each column reports performance in a held-out year using models estimated with the preceding years shown in the first row. Statistics are weighted by member-months. PMPM denotes per member per month. Mean absolute insurer-year bias is the average absolute difference between actual and predicted mean PMPM across insurers in the validation year.

D Details on Marginal Cost Residual Estimation

This appendix describes the two steps used to recover the residual component of marginal cost. I first invert residual marginal cost from observed prices for county-firm-year cells in which the firm is active. Because the medical-loss-ratio (MLR) constraint may be either slack or binding, the inversion constructs a candidate under each regime and retains a cost only when the data select a unique admissible candidate. I then project the retained costs onto observed firm and market characteristics. The fitted projection supplies residual-cost predictions for firm-market cells without a uniquely inverted cost, while its residuals form the empirical distribution of the post-entry marginal-cost shock.

D.1 Marginal-cost inversion

Let $Q_{fm}(p)$ denote enrollment, $R_{fm}(p)$ premium revenue, and $H_{fm}(p)$ expected claims expenditure for firm f in county-year market m, evaluated at the observed prices. The demand system determines these objects and their derivatives with respect to the firm's own price. I write the residual component of marginal cost as c_{fm}, so variable profit is

$$\pi_{fm}(p, c_{fm}) = R_{fm}(p) - H_{fm}(p) - c_{fm}Q_{fm}(p). \tag{3}$$

The hard MLR constraint requires claims plus residual costs to be at least 80 percent of premium revenue. Defining

$$g_{fm}(p, c_{fm}) = H_{fm}(p) + c_{fm}Q_{fm}(p) - \tau R_{fm}(p), \qquad \tau = 0.80, \tag{4}$$

the constraint is $g_{fm}(p, c_{fm}) \geq 0$.

Because the MLR regime is not observed, I construct two residual-cost candidates for every active county-firm-year cell. Under a slack MLR constraint, the multiplier is zero and the candidate c_{fm}^S solves the ordinary pricing first-order condition,

$$\frac{\partial \pi_{fm}(p, c_{fm}^S)}{\partial p_{fm}} = 0, \qquad g_{fm}(p, c_{fm}^S) \geq 0. \tag{5}$$

Under a binding constraint, the candidate is obtained directly from the MLR equality,

$$c_{fm}^B = \frac{\tau R_{fm}(p) - H_{fm}(p)}{Q_{fm}(p)}. \tag{6}$$

I then recover the implied multiplier λ_{fm} from

$$\frac{\partial \pi_{fm}(p, c_{fm}^B)}{\partial p_{fm}} + \lambda_{fm} \frac{\partial g_{fm}(p, c_{fm}^B)}{\partial p_{fm}} = 0 \tag{7}$$

and require $\lambda_{fm} \geq 0$. Thus, each candidate must satisfy the restrictions associated with the regime under which it was constructed. I retain the cell only when exactly one distinct candidate remains.

The inversion begins with 434 observed active county-firm-year cells. Constructing two regime-specific values per cell yields 868 candidates, of which 782 satisfy the strict positive-total-cost condition. The procedure retains a unique admissible residual cost in 428 cells.

D.2 Projection and residual-cost shocks

The second step maps the 428 uniquely inverted costs into residual-cost predictions for the full set of potential firm-market cells and estimates the distribution of the marginal-cost shock. While the inverted $\widehat{c}_{fm}$ all imply positive total marginal cost at the observed prices, they are not constrained to be strictly positive. As a result, if the expected cost component is sufficiently close to zero at

some counterfactual prices, the implied total marginal cost could turn negative, which is hard to interpret. For this reason, I instead estimate

$$\log\left(\widehat{c}_{fm} + h_{fm}^{\min}\right) = X'_{fm}\gamma + \omega_{fm}. \tag{8}$$

The transformed dependent variable is the logarithm of the smallest total marginal cost that the firm can face among all the possible type-plan cells in a market. Mapping a fitted value or cost shock back according to

$$c_{fm}(\omega) = \exp\left(X'_{fm}\widehat{\gamma} + \omega\right) - h_{fm}^{\min} \tag{9}$$

guarantees $hc_{ijm} + c_{fm}(\omega) > 0$ for every support-available type and plan because $hc_{ijm} \geq h_{fm}^{\min}$. The estimates shock is given by

$$\widehat{\omega}_{fm} = \log\left(\widehat{c}_{fm} + h_{fm}^{\min}\right) - X'_{fm}\widehat{\gamma}. \tag{10}$$

The covariates in X_{fm} capture prior market presence and firm-specific cost and demand conditions. They include an indicator for the firm's presence in the same county in the preceding year and the share of other counties in the same rating area in which the firm was present in the preceding year. The remaining continuous covariates summarize the firm's silver-plan primitives in the county: the mean, standard deviation, and 90th percentile of expected claims and the mean price-independent demand utility. These covariates capture the fact that residual marginal costs may vary systematically with the level and dispersion of the population's expected medical needs and with the firm's underlying demand conditions.

Table 10 reports the projection estimates. Because the dependent variable is a residual term, the individual coefficients are hard to interpret. Overall, the continuous regressors and fixed effects jointly explain 83.9 percent of the variation in the transformed residual-cost margin.

Figure 6 plots the resulting distribution of the marginal cost shocks. The distribution is centered close to zero with most observations concentrated in a relatively narrow interval. These residuals form the empirical marginal-cost-shock distribution used by the subsequent cost-draw construction.

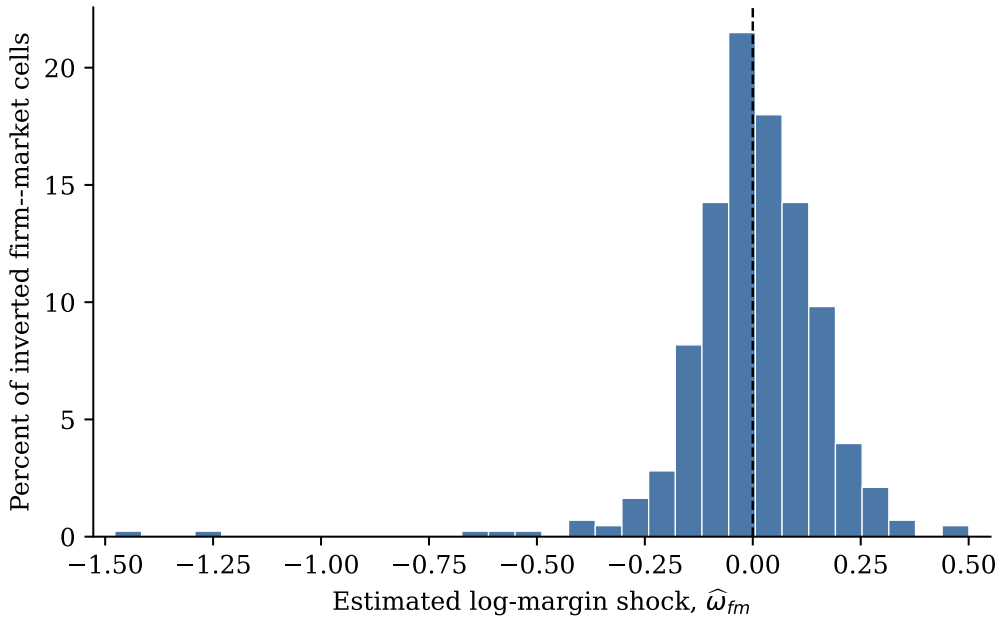
To summarize total marginal cost at the same firm–market level, I first average hc_{ijm} across the

Table 10: Projection of the Residual-Cost Margin

	$\log(\widehat{c}_{fm} + h_{fm}^{\min})$
Lagged presence in the same county	-0.1119^{**}
	(0.0555)
Lagged presence share in other rating-area counties	-0.0486
	(0.0472)
Mean silver-plan expected claims	-0.00103
	(0.00080)
Standard deviation of silver-plan expected claims	-0.00034
	(0.00025)
90th percentile of silver-plan expected claims	-0.00006
	(0.00018)
Mean silver-plan price-independent utility	0.0815^{***}
	(0.0095)
Firm fixed effects	Yes
County fixed effects	Yes
Year fixed effects	Yes
Observations	428
R^2	0.839
Adjusted R^2	0.805

Notes: The table reports unweighted OLS estimates. Standard errors are in parentheses. Expected-claims and utility summaries are computed over the firm's support-available silver plans and consumer types in the county. $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 6: Distribution of Estimated Residual-Cost Shocks



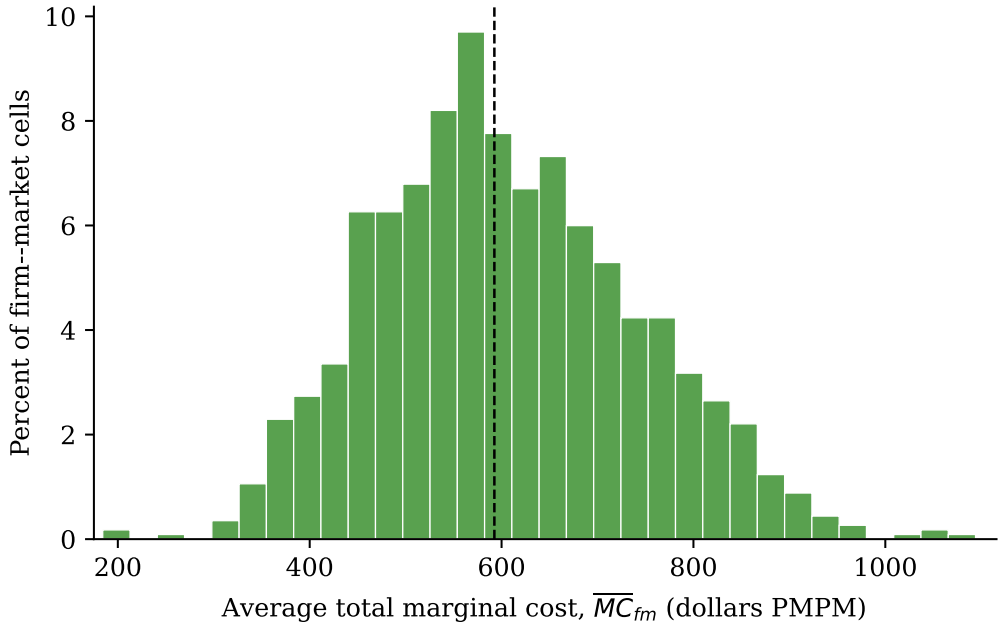
Notes: The histogram plots the residuals from equation (8). The dashed vertical line marks zero.

firm's support-available plans for each type and then average across types using their population weights. Denoting this average by $\overline{hc}_{fm}$, the cell-level measure is

$$\overline{MC}_{fm} = \overline{hc}_{fm} + \widehat{c}_{fm}. \tag{11}$$

Figure 7 displays the distribution of $\overline{MC}_{fm}$. The median is approximately \$592 per member per month, and the 99th percentile is approximately \$925.

Figure 7: Distribution of Average Total Marginal Cost Across Firm-Market Cells



Notes: For each firm-market cell, predicted claims are averaged across the firm's support-available plans within type and then across types using population weights. The final cell-specific residual-cost estimate is added to this average. The dashed vertical line marks the median.

E Entry Estimation Details

The estimation of the fixed cost parameters proceeds in two stages. The first stage estimates firms' entry policy function played in the observed equilibrium and uses them to calculate expected variable profits over possible rival-entry configurations. The fixed-cost parameters are then estimated using maximum likelihood and GMM in the second stage. This appendix section explains the implementation of each stage.

E.1 First stage: entry policy function estimation

To estimate firms' entry policy function played in the observed equilibrium, one needs to determine (1) which economic states affect payoffs, and (2) how firms map these states into entry decisions. While one could directly estimate the mapping between a large number of unstructured market states and the observed entry decisions in principle, in practice, the much smaller set of binary entry data is unlikely to learn both (1) and (2).

Because equilibrium entry probabilities are a function of own and rival profits, market states that predict own and rival profits should also determine entry probabilities. This observation motivates using the large number of pre-solved stage-game profits to learn the market states predicting profit. The implementation begins with a large collection of insurer and market characteristics. The candidate library contains 3,326 insurer–county–year features. These variables summarize characteristics that may help predict entry, including the firm's relative attractiveness, its predicted cost position, market support, network-related measures, comparisons with the closest competitors, and summaries of rivals' lagged market presence.

To select from this large set of variables, I train a neural network to screen the candidate variables for their ability to predict pricing-stage profits. The training data contain one observation for each focal insurer–county–year and each of the 32 possible configurations of the other five firms. Five indicators for the rival-entry configuration enter the neural network as context controls but cannot be selected for the final policy function. Inclusion of these indicators allows the neural network to learn how the same underlying market characteristics predict profits under different competitive environments, including interactions between market characteristics and rival entry. Then, a cross-validation procedure adds the neural screen's 40 highest-ranked variables sequentially to an L2-penalized Logit model. [16] It tests every remaining candidate and adds the one producing the greatest improvement in cross-validated log loss until the reduction in log loss is less than 0.005. This retains 12 predictors.

After selection, I augment the policy function with the focal firm's lagged same-county presence, yielding 13 predictors, and re-estimate the policy function on the full sample using an unpenalized Logit. This binary variable is a flexible first-stage predictor; it is not the fixed-cost shifter in

[16] The neural network assigns a learnable sparsity gate to each candidate characteristic.

the structural second stage, which uses continuous lagged enrollment share. The resulting fitted probabilities provide the estimated equilibrium entry policies used to form beliefs about rivals' entry when calculating expected variable profits in the second stage.

E.2 Second stage: estimation of the fixed cost parameters

Index a firm–county–year observation by $i = (f, m, t)$, and let Y_i denote the observed entry decision. For each of the $2^5 = 32$ configurations $a_{-f} \in \{0,1\}^5$ of the other five potential insurers, define

$$\overline{V}_i(a_{-f}) \equiv \mathbb{E}_\omega \left[V_{fmt}(1, a_{-f}, \omega_{mt}) \right],$$

the focal firm's pricing-stage variable profit when it enters and its rivals' entry decisions are a_{-f}, averaged over the marginal-cost draws. If $\widehat{q}_{jmt}$ is rival j's fitted first-stage entry probability, the policy-weighted expected variable profit used in the second stage is

$$\widehat{V}_i = \sum_{a_{-f} \in \{0,1\}^5} \overline{V}_i(a_{-f}) \prod_{j \neq f} \widehat{q}_{jmt}^{a_j} (1 - \widehat{q}_{jmt})^{1 - a_j}. \tag{12}$$

Thus, the first stage is used only to weight rival-entry configurations; the profit associated with each configuration comes from the solved pricing game.

Let

$$S_i = \frac{E_{fm,t-1}}{\sum_{k \in F} E_{km,t-1}}$$

be firm f's share of the six modeled insurers' total unconditional enrollment in county m in year $t-1$. The second-stage fixed-cost specification is

$$FC_i = \mu + \alpha S_i + \eta_i, \qquad \eta_i \sim N(0, \sigma_\nu^2),$$

so that, for $\theta = (\mu, \alpha, \sigma_\nu)$,

$$p_i(\theta) = \Phi\left(\frac{\widehat{V}_i - \mu - \alpha S_i}{\sigma_\nu} \right). \tag{13}$$

The fixed-cost equation contains a constant and the continuous lagged-enrollment share. It does not include a lagged-presence indicator, rating-area fixed effects, or market-size scaling.

Maximum likelihood. The maximum-likelihood estimator maximizes the Bernoulli log likelihood

$$\ell_N(\theta) = \sum_{i=1}^{N} \{Y_i \log p_i(\theta) + (1 - Y_i) \log[1 - p_i(\theta)]\}.$$

GMM. The GMM estimator uses payoff summaries constructed without the fitted first-stage entry probabilities. First, define the equal-weight payoff level

$$P_i = \frac{1}{32} \sum_{a_{-f} \in \{0,1\}^5} \overline{V}_i(a_{-f}).$$

Second, define the average loss in the focal firm's profit from adding one rival,

$$C_i = \frac{1}{5 \times 16} \sum_{j \neq f} \sum_{a_{-(f,j)} \in \{0,1\}^4} \left[\overline{V}_i(a_j = 0, a_{-(f,j)}) - \overline{V}_i(a_j = 1, a_{-(f,j)}) \right]. \tag{14}$$

Thus, P_i averages all 32 rival states with weight $1/32$, while C_i averages the 80 absent-versus-present comparisons generated by five rivals and 16 configurations of the remaining four rivals.

I standardize the two payoff summaries over the full estimation sample using

$$\widetilde{X}_i = \frac{X_i - \overline{X}}{s_X}, \qquad s_X^2 = \frac{1}{N} \sum_{i=1}^{N} (X_i - \overline{X})^2, \qquad X \in \{P, C\},$$

and leave S_i in levels. With

$$h_i = (1, S_i, \widetilde{P}_i, \widetilde{C}_i)',$$

the four sample moments are

$$g_N(\theta) = \frac{1}{N} \sum_{i=1}^{N} h_i \left[Y_i - p_i(\theta) \right]. \tag{15}$$

The reported GMM estimator minimizes the one-step, identity-weighted criterion

$$\widehat{\theta}_{\text{GMM}} = \arg\min_{\theta} g_N(\theta)' g_N(\theta).$$

These four moments require the entry residual to have zero sample mean and to be orthogonal to lagged enrollment share, the equal-weight payoff level, and the average profit loss from additional competition. The constant and share moments discipline average entry and its relationship

with prior enrollment; the two payoff moments supply variation in the level and competitive sensitivity of profits. There are four moments for the three parameters, leaving one overidentifying restriction. Unlike $\widehat{V}_i$, P_i and C_i use equal configuration weights and therefore do not depend on the fitted conditional choice probabilities.

Inference. The reported standard errors are sandwich standard errors clustered by county. They are conditional on the fitted first-stage entry probabilities and the precomputed pricing-stage payoff schedule.

F Equilibrium Solve Details

This appendix describes the algorithm used to solve the pricing equilibrium for each county-year market, entrant set, and marginal-cost draw. The procedure has four main steps. First, it constructs an initial price vector from a previously solved, economically similar case. Second, it searches for a solution to the firms' exact first-order and hard medical-loss-ratio (MLR) conditions. Third, if the exact search fails, it uses coordinate-wise best responses to find an economically stable price vector. Finally, it removes firms with fewer than one expected enrollee and repeats the pricing calculation for the remaining firms. A result is retained only if it satisfies either the exact constrained first-order conditions or the best-response acceptance criteria described below.

F.1 Pricing problem recap

Let $Q_{fm}(p)$, $R_{fm}(p)$, and $H_{fm}(p)$ denote enrollment, gross premium revenue, and expected claims for firm f in county-year market m. Let c_{fm} denote its residual marginal cost. Profit is

$$\pi_{fm}(p) = R_{fm}(p) - H_{fm}(p) - c_{fm}Q_{fm}(p). \tag{16}$$

The MLR constraint is

$$g_{fm}(p) = H_{fm}(p) + c_{fm}Q_{fm}(p) - 0.80R_{fm}(p) \geq 0. \tag{17}$$

F.2 Constrained first-order conditions

For each active firm, the solver considers whether the MLR constraint is slack or binding. If it is slack, the firm's own-price profit derivative must equal zero and the constraint must remain feasible:

$$\frac{\partial \pi_{fm}(p)}{\partial p_{fm}} = 0, \qquad g_{fm}(p) \geq 0. \tag{18}$$

If the constraint binds, the solver requires

$$g_{fm}(p) = 0, \qquad \lambda_{fm} = -\frac{\partial \pi_{fm}(p)/\partial p_{fm}}{\partial g_{fm}(p)/\partial p_{fm}} \geq 0. \tag{19}$$

F.3 Initial values and solution order

I first solve each entrant set with no marginal cost shock as a warm start value for the later cases with different marginal cost shocks. Then, for each entry set, the marginal-cost draws are ordered so that successive cost vectors are close. Each draw begins from a previously solved draw for the same entry set.

F.4 Constrained first-order solution search

The solver first checks whether the initial prices already satisfy the constrained first-order conditions. It then tests the binding set inherited from the warm-start source. If that binding set yields an exact solution, the search ends. Otherwise, the solver constructs a numerical warm start by solving smoothed Fischer–Burmeister complementarity equations. [17]

If none of the warm-start binding sets yield a solution, the solver then enumerates the remain-

[17] For each active firm, the least-squares solver jointly chooses prices and MLR multipliers to minimize the residuals from the stationarity equation

$$\frac{\partial \pi_{fm}(p)}{\partial p_{fm}} + \lambda_{fm} \frac{\partial g_{fm}(p)}{\partial p_{fm}} = 0$$

and the Fischer–Burmeister equation

$$\phi\Big(\lambda_{fm}, g_{fm}(p)\Big) = \sqrt{\lambda_{fm}^2 + g_{fm}(p)^2} - \lambda_{fm} - g_{fm}(p) = 0.$$

The Fischer–Burmeister equation holds if and only if $\lambda_{fm} \geq 0$, $g_{fm}(p) \geq 0$, and $\lambda_{fm} g_{fm}(p) = 0$, thereby imposing MLR feasibility and complementary slackness without specifying in advance whether each firm's constraint binds. Numerically, the solver replaces ϕ with the smooth approximation

$$\phi_\tau\Big(\lambda_{fm}, g_{fm}(p)\Big) = \sqrt{\lambda_{fm}^2 + g_{fm}(p)^2 + \tau^2} - \lambda_{fm} - g_{fm}(p)$$

and progressively reduces τ to approach the discrete solution.

ing possible sets of binding MLR constraints among the active firms, ordered by (1) increasing number of binding firms; and (2) firm ID within sets of the same size. For every candidate binding set, the solver first tries to find prices that solve the corresponding system of binding MLR equations and slack first-order conditions. After optimization, the solver verifies the slackness classification by checking MLR feasibility, slack-firm profit derivatives, binding-firm MLR gaps, implied MLR multipliers, and constraint multiplier nonnegativity. The candidate is accepted only if the maximum violation (normalized by enrollment) across all these conditions is at most 0.001. The solver stops at the first valid binding set.

F.5 Best-response resolution

If no binding set produces an acceptable hard-KKT solution, the solver searches for profitable unilateral price deviations. A deviation is economically material when its profit gain exceeds

$$\min\{\$100,\ 0.01\max(0, \text{current profit})\}. \tag{20}$$

When a material profitable deviation exists, the solver updates the firm with the largest feasible gain and repeats. A result is classified as solved with a best-response resolution if every firm is best responding to the other firms with a feasible price and no firm has a material profitable deviation.

F.6 Pruning

After a pricing solution is obtained, all active county-firm cells with expected enrollment strictly below one are removed simultaneously. The reduced entry set is then solved again, starting from the preceding prices, until all the remaining firms reach an acceptible solution.